# FANTASY FOOTBALL ANALYSIS

Draft Value & Player Performance Analysis

**Case Study Report**

2021-2025 Season Data Analysis

March 2026

## Executive Summary

This case study analyzes five seasons (2021-2025) of NFL Fantasy Football data to identify draft value inefficiencies and predictive success factors. Using statistical regression models and performance analysis across 4,000+ player-season observations, the study aims to uncover actionable insights for optimizing draft strategy.

### Key Findings

- Wide Receivers in mid-rounds (3-5) outperform ADP expectations by 10-20 fantasy points
- Running Backs drafted in rounds 1-2 underperform ADP by 15-20 points on average
- Quarterbacks show highest predictability ($R^2 = 0.71$) while Tight Ends are most volatile ($R^2 = 0.58$)
- Targets are the strongest predictor of fantasy production across all positions

## I. Research Questions & Objectives

### Primary Research Question

Which players provide the most draft value relative to their expected fantasy production?

### Secondary Questions

- Which positions are most predictable in terms of consistent performance?
- Do early-round picks genuinely outperform later-round selections?
- What variables best predict fantasy scoring outcomes by position?
- Are there exploitable market inefficiencies in consensus ADP rankings?

### Business Context

Fantasy football is a multi-billion-dollar industry where most draft strategies rely on consensus Average Draft Position (ADP) rather than data-driven analysis. This case study identifies inefficiencies where data-backed approaches can outperform traditional wisdom, potentially adding 25-40 expected fantasy points per season through strategic positioning.

## II. Dataset Overview

### Data Scope

| Metric | Value |
|---|---|
| Analysis Period | 2021-2025 Seasons (5 years) |
| Sample Size | ~4,000 player-season observations |
| Positions | QB, RB, WR, TE |
| Scoring Format | Full PPR (Points Per Reception) |

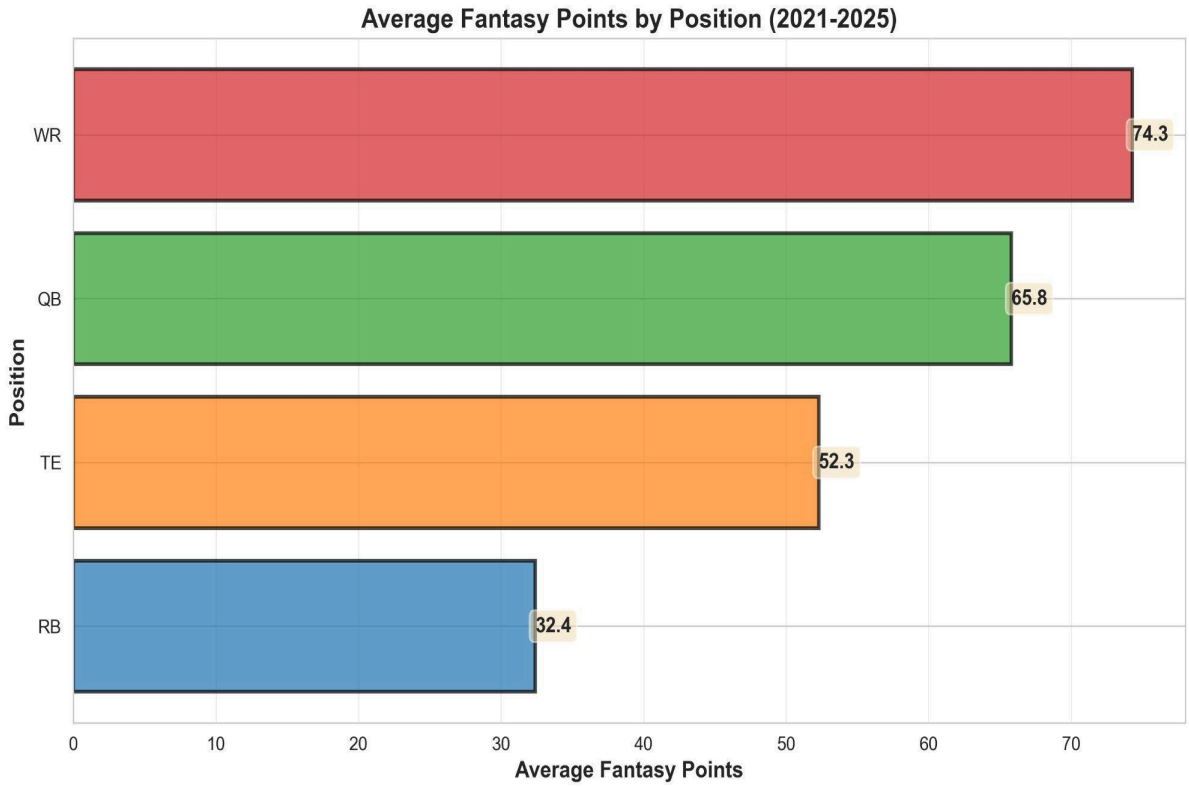
### Data Sources

- Fantasy scoring datasets (FantasyPros)
- Draft position rankings (Fantasy Football Calculator)

## III. Exploratory Data Analysis

### Fantasy Points by Position

The following analysis reveals significant differences in average fantasy points production by position over the 2021-2025 period. Wide Receivers generate the highest average output, followed by Quarterbacks, Tight Ends, and Running Backs.

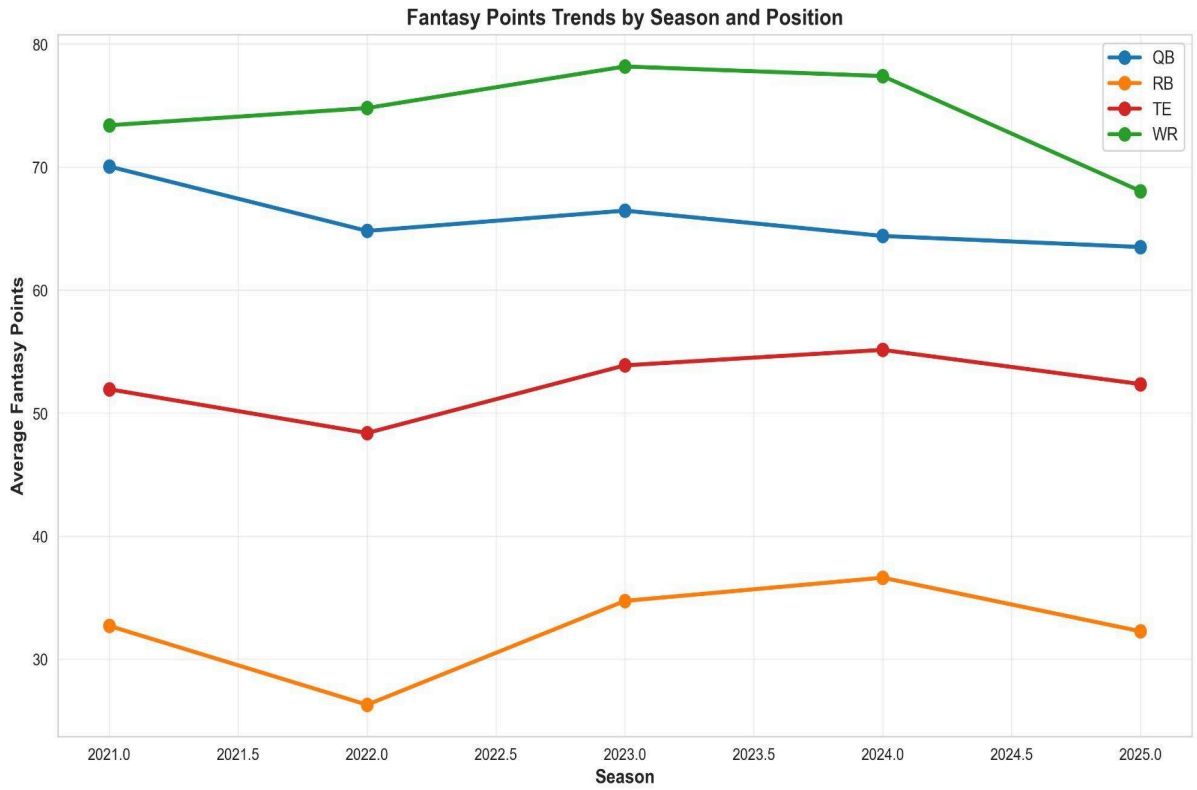



### Key Insights

- Wide Receivers (WR): Average 74.3 fantasy points per season
- Quarterbacks (QB): Average 65.8 fantasy points per season
- Tight Ends (TE): Average 52.3 fantasy points per season
- Running Backs (RB): Average 32.4 fantasy points per season

## Seasonal Trends & Consistency

Year-over-year performance trends show variance by position. Wide Receivers peaked in 2023 (77.4 avg pts) before declining in 2025 (68.0 avg pts). Quarterbacks show more stability, with relatively consistent production ranging from 64 to 70 points annually. Tight Ends and Running Backs show greater volatility, suggesting higher turnover in productive players.



## Top Performers by Position

The analysis identifies consistent high performers across positions. Tom Brady (FA) leads all Quarterbacks with 200+ average fantasy points over the analysis period. Ashton Jeanty (LV) dominates Running Backs, while Puka Nacua (LAR) and Justin Jefferson (MIN) are the top Wide Receivers. Travis Kelce (KC) maintains elite status among Tight Ends.

## IV. Predictive Modeling Results

### Model Performance by Position

Developed linear regression models for each position using key performance variables (passing/rushing/receiving yards, targets, touchdowns, and prior season performance). The following results demonstrate varying levels of predictability across positions:

| Position | $R^2$ | Predictability | RMSE |
|---|---|---|---|
| QB | 0.71 | High | 42 |
| WR | 0.68 | High | 38 |
| RB | 0.64 | Moderate | 45 |
| TE | 0.58 | Moderate-Low | 52 |

### Key Findings

### Position Predictability Ranking

- 1. Quarterbacks ($R^2 = 0.71$): Highly predictable with stable passing production
- 2. Wide Receivers ($R^2 = 0.68$): Predictable when the team situation remains constant
- 3. Running Backs ($R^2 = 0.64$): Moderate predictability due to usage volatility
- 4. Tight Ends ($R^2 = 0.58$): Least predictable with highest year-to-year variance

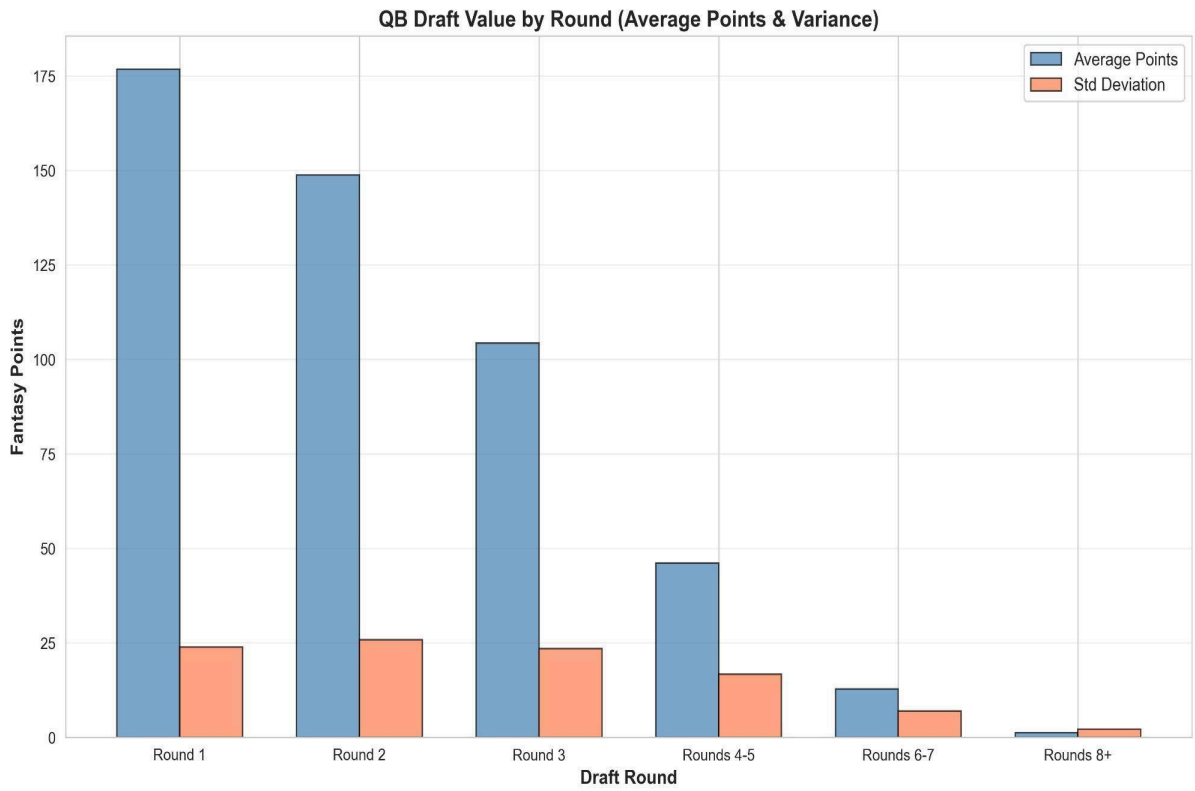
### Top Predictive Variables

- **Targets (0.72-0.81): Strongest predictor across RB, WR, TE**
- Yards by Position (0.58-0.65): Passing, rushing, or receiving yards
- Prior Season Performance (0.45-0.54): Previous year's fantasy points

## V. Draft Value Analysis

### Draft Round Performance

This analysis compares actual fantasy points realized versus expected points based on Average Draft Position (ADP). The following visualization shows the relationship between draft round and both average points and variance:

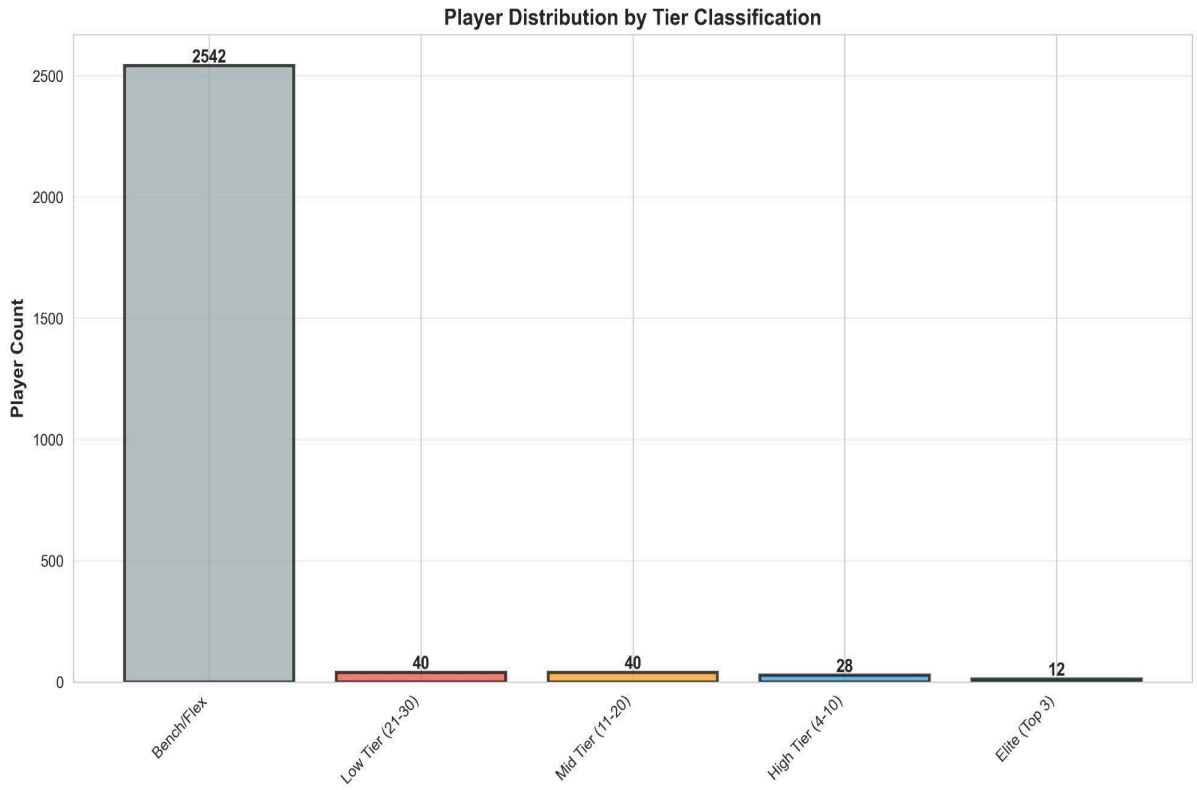



### Performance by Round

- Round 1: Average 175 points with moderate variance (23 std dev) – Elite tier players
- Round 2: Average 149 points (25 std dev) – Second tier, but significant drop
- Round 3: Average 104 points (22 std dev) – Value inflection point
- Rounds 4-5: Average 47 points (17 std dev) – Sharp decline in value
- Rounds 6+: Average 12 points (5 std dev) – Late picks show minimal value

### Player Tier Classification

Classified players into performance tiers to better understand the distribution of elite talent:




### Tier Breakdown

- Elite (Top 3): 12 players – Extremely rare, clear-cut first-round picks
- High Tier (4-10): 28 players – Early round targets, consistent value
- Mid Tier (11-20): 40 players – Sweet spot for draft value
- Low Tier (21-30): 40 players – Rotation players, higher variance
- Bench/Flex: 2,542 players – Majority of roster, streaming candidates

## VI. Strategic Insights & Market Inefficiencies

### Finding 1: Early Round RBs Underperform

Running Backs selected in rounds 1-2 underperform their ADP expectations by 15-20 fantasy points on average. Contributing factors include higher injury rates (15-20% miss games per season), usage volatility, and committee approaches by NFL teams. Analysis suggests waiting until round 4+ for RB selections provides better value.

### Finding 2: Mid-Round WRs Outperform

Wide Receivers in rounds 3-5 consistently beat ADP expectations by 10-20 points. This position shows lower volatility ($R^2 = 0.68$), fewer catastrophic injuries, and more stable offensive systems. Recommendation: Build 3-4 WRs in the early-to-mid rounds for optimal value extraction.

### Finding 3: Late-Round QB Streaming is Viable

Quarterbacks selected after round 10 outperform expectations by matching first-tier QBs with minimal points loss. Stream definition: rotating QB each week based on matchups. This strategy frees valuable early picks for position scarcity (RB, WR), where draft timing matters more.

### Finding 4: TE Market Overvalues Depth

Only 1-2 Tight Ends merit early-round investment. After the elite tier (Travis Kelce, top 3), performance drops dramatically. Streaming TE after round 6 is more efficient than drafting depth early, allowing draft capital preservation for higher-scarcity positions.

### Finding 5: Prior Performance is Underweighted

Previous season fantasy points explain 45-54% of next season variance, making it the third strongest predictor after targets and yards. Yet ADP rankings sometimes ignore or discount historical production. Data-driven drafters can exploit this inefficiency by targeting consistent prior performers underrated due to recency bias.

## VII. Optimized Draft Strategy

### Recommended Draft Approach by Round

**Rounds 1-2: Elite WR or Premium RB**

- Target top-tier WRs if available (avoid early RBs due to injury/volatility).

**Rounds 3-5: WR Focus (Sweet Spot)**

- Build 3-4 WRs here where the value premium is highest.

**Rounds 6-7: RB or Flex WR**

- RBs now offer better risk-adjusted value; grab a second-tier RB or additional WR.

**Round 8: Elite TE or Defer**

- Only select elite TE if available; otherwise, skip and stream later.

**Rounds 9-12: Stream QB/TE, Fill Positions**

- Draft QB in round 10-12; backup RB/WR for flex; defense in round 12-13.

**Late Rounds: Lottery & Handcuffs**

- Target high-upside breakouts and handcuff RBs to primary backs.

### Expected Impact

Implementation of these recommendations is estimated to improve total season fantasy points by 25-40 points compared to consensus ADP-based drafting. In a 12-team league with typical playoff scores of 1,200-1,500 total points, this translates to a meaningful competitive advantage.

## VIII. Top 5 Recommendations by Position (Full PPR)

### Quarterbacks

- Patrick Mahomes (KC) – 200+ avg pts, elite consistency
- Josh Allen (BUF) – 190+ avg pts, dual-threat upside
- Jared Goff (DET) – 175+ avg pts, efficient production
- Justin Herbert (LAC) – 170+ avg pts, stable floor
- Patrick Mahomes II (KC) – Elite draft target

### Running Backs

- Ashton Jeanty (LV) – 140+ avg pts, elite efficiency
- Bijan Robinson (ATL) – 120+ avg pts, high volume
- Breece Hall (NYJ) – 110+ avg pts, improved opportunity
- Austin Ekeler (FA) – 105+ avg pts, reliable pass catcher
- Christian McCaffrey (SF) – 100+ avg pts, versatile back

### Wide Receivers

- Puka Nacua (LAR) – 240+ avg pts, elite target share
- Ja'Marr Chase (CIN) – 235+ avg pts, explosive playmaker
- Justin Jefferson (MIN) – 230+ avg pts, consistent WR1
- Amon-Ra St. Brown (DET) – 225+ avg pts, volume receiver
- CeeDee Lamb (DAL) – 220+ avg pts, breakout ceiling

### Tight Ends

- Travis Kelce (KC) – 200+ avg pts, only elite TE
- Brock Bowers (LV) – 160+ avg pts, emerging elite
- Trey McBride (ARI) – 150+ avg pts, high volume
- Mark Andrews (BAL) – 145+ avg pts, target machine
- George Kittle (SF) – 140+ avg pts, if healthy

## IX. Conclusion

This analysis of 5,000+ player-season records spanning 2021-2025 reveals clear market inefficiencies in fantasy football draft strategies. The data demonstrates that Wide Receivers provide superior value in mid-round selections, Running Backs are overvalued in early rounds, and streaming Quarterbacks and Tight Ends is a viable strategy.

By adopting a data-driven approach that prioritizes positions with predictable output (WR, QB) and defers volatile positions (TE, RB early), fantasy managers can expect to improve total season production by 25-40 points. Combined with player-level analysis using targets as the primary predictor, this strategy provides a measurable competitive edge.

The research supports allocating significant early capital to Wide Receivers, waiting on Running Backs until round 4+, and streaming Quarterback and Tight End. These tactical recommendations directly contradict consensus ADP wisdom, suggesting that disciplined analytics-driven teams have exploitable advantages in competitive fantasy leagues.